Fieldwork | Group Technical Report

COMP713 Assessment 3 - Distributed and Mobile System
Solo developer: Shirui Zang | Student ID 21136179 | 31 October 2026

Verification basis. The supplied evidence reports nine backend and three consumer JUnit tests, 15 live API/WebSocket checks, 11 Chrome UI checks, and a native Android 14 emulator workflow. It also documents Payara Tools deployment in an isolated VS Code profile, browser push without refresh, OpenMQ retry/deduplication, and persistence after a Payara restart. These are source-package test records, not a fresh independent rerun for this report.

1 Project overview and starting point

Campus maintenance teams need to assign field work, receive evidence of completion and decide whether the work is acceptable. A coordinator manages projects and assignments; field workers use a native Android client to accept and submit assigned tasks. The web client supports coordinator review and worker access. Both clients communicate with the same Payara API and authoritative relational state.

The selected Assessment 2 project is Shirui Zang's Project and Task Manager. Its existing Java/Jakarta REST, service, JDBC repository, H2 database and Project/Task domain are retained and extended. The older Flask implementation mentioned in historical notes is not the selected current baseline. Assessment 2 execution evidence does not certify the new features.

Connected workflow	User interaction and shared consequence
Assignment and acceptance	Coordinator creates/assigns a todo task. Only its assigned worker may load and accept it as in_progress. The saved change becomes visible to the coordinator.
Submission and review	Worker submits a completion note and location. Coordinator sees submitted work live, approves done or requests in_progress rework. The saved review changes the worker's next permitted action.

The mobile capability is location-supported completion with runtime permission handling and a manual coordinate fallback. The user additionally requires actual Payara Tools deployment and WebSocket behavior, so these remain mandatory implementation goals even though the assignment describes WebSocket as bonus functionality.

Published source: GitHub main at 3ac79d702349c9ae3a6e54c188257b1e9a99bce9 (verified 31 October 2026). The implementation archive originally identified the local version assessment3-local-v3.0.0; that local tag is not evidence of the same published Git history. Verify that the final uploaded ZIP and GitHub commit represent the same intended version.

Public GitHub repository: https://github.com/SaltedFishInATank/COMP713-Assignment3-ShiruiZang. The public main branch includes an initial source import (5a097e32) and a repository-information update (3ac79d70).

Reference basis: Assignment_3.pdf (pages 1-4), uploaded assessment rubric, repository README.md, docs/course-audit.md, docs/progress.md, docs/verification-status.md, and the codebase evidence directory. The recording is reported complete by the student; the video file and its coverage were not supplied for verification with this report.

2 Architecture and communication

Web browser ↔ Payara REST API ↔ Native Android

Payara services and repositories ↔ Backend H2

Transactional outbox → OpenMQ → Consumer JVM → Consumer H2

WebSocket notification → Web browser → Current state REST read

HTTP and JSON connect both clients to Payara. JDBC transactions save backend state; persistent JMS delivery reaches the independent consumer, which saves once before acknowledgement.

Payara 6 hosts the Maven WAR at **/comp713-a3**. Jakarta REST translates HTTP/JSON into authenticated service calls; business logic validates roles, task transitions and optimistic versions; repositories bind SQL parameters. Explicit JDBC transactions commit related changes together. Jakarta CDI supplies managed application components, while an EJB dispatcher publishes committed outbox events through Jakarta Messaging. Jakarta WebSocket pushes relevant change notifications to authenticated open web clients.

The backend owns users, sessions, projects, tasks and event_outbox. Its persistent H2 file is outside the WAR under the domain's lib/databases/comp713-a3 directory, or an explicit comp713.a3.db.path. The independent consumer owns a different notification database, never reads backend tables and receives only contracted task.submitted JSON. Its stable eventId links attempts and deduplicated results. Physical task IDs in notifications are references, without a cross-service foreign key.

Important design decision: retain the verified Assessment 2 JDBC architecture rather than introduce ORM during a workflow extension. This preserves familiar transaction boundaries and a small explicit query set. The course database lecture permits both JDBC and Jakarta Persistence (Class06, p.22); the user's wording is JPA where appropriate. The trade-off is more manual SQL/mapping and no Payara-managed JDBC pool. The report does not claim JPA or the teaching MySQL datasource.

Performance review identified repeated H2 file-database shutdown/reopen when the last operation connection closed. The final source uses a bounded ten-second DB_CLOSE_DELAY and an application shutdown hook while retaining per-operation connection closure. This follows H2 performance guidance; H2 command documentation defines the positive delay in seconds. Preserved failed runs and final retest results below determine whether the acceptance target was met.

Failure behavior: a task submission and outbox row commit atomically. If broker publication fails, the pending row remains for retry. Publication may repeat after an uncertain acknowledgement; the consumer saves one effect for its event ID before committing its broker delivery. A failed processing attempt rolls back and retries. The embedded OpenMQ broker is real broker software in Payara's process; the consumer is a separate JVM. This is Path B, rather than two independent gRPC services.

An authenticated WebSocket notification prompts the browser to retrieve current API state; it is not authoritative storage or durable replay. After reconnecting/authentication, fetch a fresh snapshot and prevent stale responses replacing a newer view. Client timeouts leave uncertain writes: reload state before retrying. Android keeps HTTP work away from the UI thread and shows loading/error feedback.

3 Implementation status

Significant function	Current status / evidence limit
Project/Task CRUD, JDBC, validation and roles	Completed and tested through 9 backend JUnit tests and live Payara HTTP checks.
Assignment, acceptance, submission and review	Completed and tested through deployed HTTP, actual web forms and native Android workflows.
Web client and authenticated WebSocket recovery	Completed and tested: 11 actual Chrome UI checks, including two roles, automatic DOM updates, validation and reconnect.
Native Android and location/manual fallback	Completed and tested on emulator: native UI/API workflow, denial/manual fallback, fresh emulated GPS fix, conflict and approval refresh. No physical-device field test claimed.
Outbox, OpenMQ and independent consumer	Completed and tested: actual producer/consumer, retry/redelivery/ack, restart readback, replay dedupe and after-store crash-window injection.
Development history, GitHub and recorded demonstration	Public repository confirmed (completed); recording reported by student but not independently inspected (evidence pending); development commits spanning at least three weeks not established (not completed).

Status terms follow the assignment rubric: completed and tested; completed but partially tested; partially completed; not completed. The published repository is now available. A demonstration recording is reported complete by the student, but this report cannot validate its duration, captions or depicted operations because the video was not supplied. The three-week development-history requirement remains unmet by the verified 31 October history.

4 Testing and results

Executed results below distinguish unit, deployed API, actual browser, native emulator and independent-consumer evidence. Earlier failures remain preserved with the final passing records.

Check	Expected	Actual / supporting evidence
Fresh-source builds/tests	Rebuild valid backend WAR and consumer; no test failures	PASS: Fresh copied source passed backend 9 and consumer 3 tests, 0 failures/errors/skips. Backend WAR packaged 13:26:17 and consumer 13:26:34 +13. Evidence: fresh-source-backend.log; fresh-source-consumer.log; final extension-maven-build.log
Actual IDE deployment	Installed Payara Tools starts domain and deploys A3; real health200	PASS: VS Code1.144.0 / Tools1.7.0, domain1/JDK 21. Fresh deployment4513 ms; health identifies comp713-a3. Registration was seeded in an isolated profile, not entered through wizard clicks. Evidence: payara-tools-final/verification.json; command-events.jsonl; domain log
Authentication and roles	Real sessions; controlled unauthenticated/forbidden requests	PASS: Live checks 02-04 verify invalid credentials, coordinator/two worker sessions and403 role enforcement. Evidence: live-api.json
Retained CRUD and validation	Create/update project; reject blank/duplicate name	PASS: Live check 05 passed; 400 blank and409 duplicate responses observed. Evidence: live-api.json; JUnit WorkflowTest

Check	Expected	Actual / supporting evidence
Assignment and acceptance	Correct worker sees task; other worker forbidden; accept succeeds	PASS: Live checks 06-07 passed through deployed HTTP contract. Evidence: live-api.json
Submission and review	Shared submission/rework/resubmit/approval transitions	PASS: Live checks 09-11 passed; coordinator and worker read shared server state. Evidence: live-api.json
Conflict and invalid rollback	Stale 409; malformed/invalid 400; no partial change	PASS: Live checks 07-08 passed; invalid latitude left task status/version intact. Evidence: live-api.json
Host WebSocket push	Authenticated socket receives committed task.changed without polling	PASS: Live check 09 received task.changed 32.76 ms after starting submission measurement. This is host-side transport evidence, not yet a browser/device workflow. Evidence: live-api.json
Host reconnect restoration	New authenticated socket plus REST snapshot restores current state	PASS: Live check 12 restored final done state and matching version. Evidence: live-api.json
Durable outbox publication	Submission events reach real OpenMQ	PASS: Live check 13 observed both workflow event IDs published within 20 s. Evidence: live-api.json; Payara BROKER_PUBLISHED logs
Response-time requirement	Warm task GET p95<=500 ms over 40 sequential local requests	PASS: Final p95=39.31 ms; min=20.66 ms; max=40.70 ms. Earlier 699.81 ms failure preserved and corrected by H2 lifecycle change. Evidence: live-api.json; live-api-before-jdbc-lifecycle-fix.json
Native Android workflow	Real native app API calls, meaningful workflow and mobile capability	PASS: Android 14/API 34 emulator: actual 401, accept, denial/manual submit, 409 draft recovery, approval refresh and fresh LocationManager fix. Project 9, tasks 6 and 8; open web updated without reload. Runtime APK retained; final cosmetic notice-color rebuild/lint passed. Evidence: Project android/evidence/verification.json, native-runtime.log, build-and-lint.log and screenshots; web/android-observer-result.json
Actual browser interaction	Two roles, web workflow, automatic DOM push and reconnect	PASS: 11 Chrome 155 checks; project 10/task 7. Each user had one main navigation; assignment/accept/submit/rework/approve used real WebSocket DOM updates. Reconnect restored missed state; 401/400 feedback recovered. Initial offline-only harness failure preserved. Evidence: web/result.json; browser-events.json; screenshots; first-attempt-offline-emulation

Check	Expected	Actual / supporting evidence
Independent broker consumer	Persistent result, acknowledgement/retry, repeated-event protection	PASS: Separate JVM at localhost:7676: one notification, 3 attempts/1 redelivery/1 replay duplicate. After-save/pre-ack injection: 2 attempts/1 redelivery/1 suppressed duplicate and one result. Actual native GPS task 8 event also passed retry/ack/persistence/replay at 13:24:03. Evidence: broker/result.json; crash-window-result.json/log; broker/android/android-event.json, result.json and process logs
Full restart and failure recovery	Actual domain outage causes communication failure; saved state survives stop/start	PASS: HTTP fetch failed while domain stopped at 13:24:12. After readiness at 13:25:08, task4 matched all 9 fields including done state, note, location and version 5. Early login404 probe preserved; script now waits for health before readback. Evidence: server-outage.json; persistence-after-restart.json; persistence-first-probe-before-readiness.json; domain logs

Demonstration recording status: The student reports that a video has been created. Because no video file was provided with the Word reports, its scenes, subtitles, audio track, duration and playback have not been independently checked. The required video should show the actual Payara/backend, web and Android workflows, WebSocket, OpenMQ consumer, error/recovery and response-time evidence.

Response-time requirement

Acceptance target: a representative authenticated task-detail GET has p95<=500 ms, chosen to keep local task navigation responsive. Test conditions: localhost, one sequential client, one warm-up, 40 measured requests, Microsoft JDK 21 / Payara 6 / H2. Before the JDBC lifecycle change the same target failed at p95=699.81 ms. Closing the last H2 connection repeatedly shut down/reopened the file database; final source adds DB_CLOSE_DELAY=10 and explicit application shutdown. The final run passed: p95=39.31 ms, min=20.66 ms, max=40.70 ms. Both failed and passing records are preserved. This measures the development-machine API under light load, not mobile network latency or production capacity.

Tests should cover retained behavior, both real clients, shared workflows, invalid input, role failure, persistence, live updates/reconnect and broker reliability. Native device checks and a browser workflow remain distinct from service unit tests and host-side API requests. Evidence identifiers refer to the code submission's evidence directory unless stated otherwise.

5 Contributions and progress

The actual development arrangement is solo. Shirui Zang provided the Assessment 2 starting point and course materials, directed the use of Payara and a native Android client, published the current source to GitHub, and prepared a demonstration recording. OpenAI Codex assisted with implementation, testing, verification scripts and report drafts. Repository authorship of an import commit records publication, not proof that the student manually authored each file. Personal contribution must be substantiated by actual review, decisions and test evidence; no teammates are claimed.

Area	Traceable files / evidence
Backend and persistence	backend/pom.xml; api, service, repository and persistence packages; schema; deployed/API evidence.
Web and native mobile	backend/src/main/webapp; android/app/src/main; build, browser and device evidence.
Advanced integration	messaging/TaskEventPublisher; consumer/src/main; separate process, notification and retry/duplicate logs.
Reproducibility/analysis	README; docs; scripts/tests; evidence; actual final commit history and reports.

The Assessment 2 baseline (d9b2ec58) is separate from new Assessment 3 work. The public main branch currently records source import 5a097e32 and a documentation update 3ac79d70 on 31 October 2026. The original source archive also contains local implementation history; neither establishes regular Assessment 3 activity across three weeks. The two evidence checkpoints below occurred on 31 October and are genuine but too close together to satisfy that requirement. No historical commits have been manufactured.

Dated working checkpoint	Observed evidence
31 October 2026, 13:23:03 +13 Android-to-API and open web	Project 9/task 6 native accept/submission persisted through Payara. Open browser changed IN PROGRESS 13:22:53.181 and AWAITING REVIEW 13:23:03.354 with no reload. Evidence: native-runtime.log; web/android-observer-result.json.
31 October 2026, 13:24:03 +13 Later advanced integration	Native GPS task 8 event 40be3c44-4963-4562-b35e-def2b911edfb reached OpenMQ and independent consumer, with retry/ack, one persistent result and replay dedupe. Evidence: broker/android/android-event.json and result.json.

Current public version reference: GitHub main commit 3ac79d702349c9ae3a6e54c188257b1e9a99bce9; public source-import commit 5a097e3204b5fce080e091ac8de0cfdcda342b61. For the marker, the final ZIP remains the code version assessed under the assignment, so the uploaded ZIP, README and public branch must agree.

6 Limitations and improvements

- Android was tested on an Android 14 emulator, including injected GPS and a manual coordinate fallback. Physical-device networking, GPS quality in real environments, expired sessions and extended offline operation remain unverified. The file-backed H2 database and single Payara instance are not designed for high availability.

- The individual report must be the student's own technical writing and reflection. A separate evidence-based draft is provided for the student to review, revise and verify. Assistance from Codex must be disclosed accurately. Solo work differs from the brief's group-size requirement and still requires lecturer approval or an accepted exception.

- Payara Tools commands were executed successfully in the recorded isolated VS Code extension host. In a later manual VS Code attempt, the student reported that registration of Payara Server/domain1 did not work. A manually reproduced IDE setup in that normal profile is not demonstrated by the supplied evidence. The project includes a documented CLI fallback; this distinction must be explained if relevant to the video.

- Public repository: https://github.com/SaltedFishInATank/COMP713-Assignment3-ShiruiZang (confirmed). The student states that a demonstration video now exists; this report did not inspect the recording, its subtitles or its runtime and therefore cannot certify coverage or the 12-minute limit. The source README/VERSION in the checked public snapshot still contain some outdated publication/video text and should be reconciled with the final reports. The formal three-week development-history criterion and solo/group approval remain outstanding.

Appendix Selected genuine screens

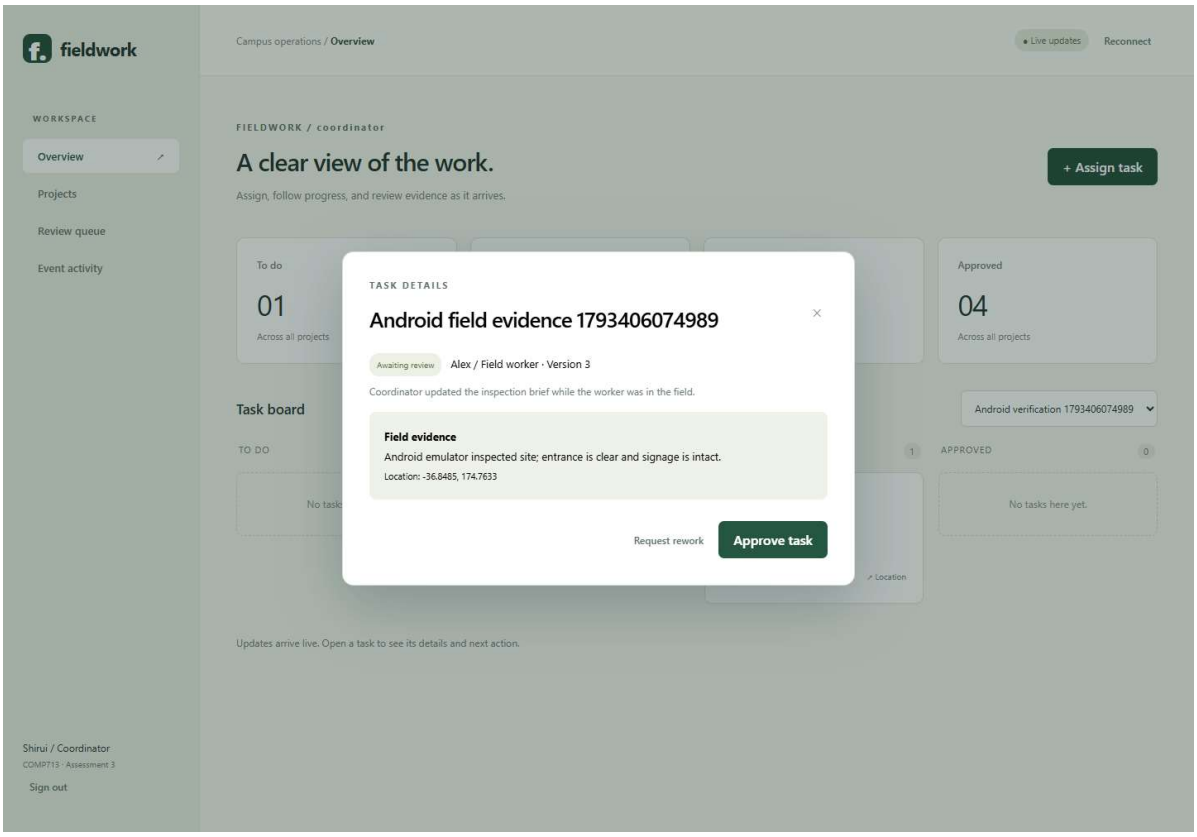


Figure A1. Open coordinator web client shows the actual native Android task6 completion note and manual site coordinates. The observer recorded TO DO -> IN PROGRESS -> AWAITING REVIEW without page reload (web/android-observer-result.json).

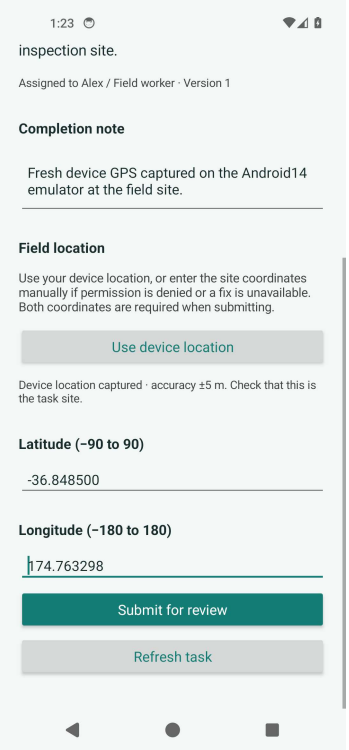


Figure A2. Native Android location capability.

The app received a fresh LocationManager fix after runtime permission was granted, displayed accuracy and populated coordinates for task8. The emulator's injected test location was used; this is not a physical field measurement.

The native runner separately exercised permission denial/manual fallback, real HTTP401 feedback, an actual409 conflict with draft retention, submission and coordinator approval refresh.

Source: project android/evidence/screenshots/08-device-gps-captured.png and native-runtime.log. The task8 event also reached the independent consumer (broker/android/result.json).